Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency

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Abstract

Large language models (LMs) store vast factual knowledge, yet turning it into a reliable, *set-valued* knowledge base, where a subject may have zero, one, or many objects, remains hard. The LM-KBC 2026 shared task poses this in a strict *closed-book* setting: an open-weight model of at most 32B parameters, with no retrieval, no external knowledge, and no fine-tuning. We start from the metric and observe that it rewards *abstention*: since an empty prediction scores a perfect F_1 against an empty gold set, predicting nothing everywhere already reaches 0.169 relation-averaged macro-F_1, and on the null-heavy relations even beats the official baseline. We therefore treat *when to answer* as a first-class decision and build a per-relation pipeline that grounds each prompt in the relation's exact scope, decides abstention from the sampled set rather than by instruction, and takes a median over samples for numeric relations. Our **Gemma-4-31B** system (30.7B, the strongest in-budget model we could access) scores $0.461/0.488$ (relation-averaged/micro) from a single sample, beating the official baseline on five of the six relations, while a smaller 4-bit Qwen2.5-14B, even with self-consistency voting, reaches only 0.416: in this comparison a larger single-sample model outscores a smaller voted one. Generation is cached, so decoding reproduces without a GPU. We release all code, prompts, and cached generations.[1]

1 Introduction

Pre-trained language models (LMs) store a remarkable amount of factual knowledge in their parameters (Petroni et al., 2019; Roberts et al., 2020), and a body of work probes and extracts it (Jiang et al., 2020; Zhong et al., 2021; AlKhamissi et al., 2022). This raises the prospect of building a structured knowledge base (KB) directly from an LM rather than from curated sources (Dong et al., 2014; Vrandečić and Krötzsch, 2014). Doing so is harder than answering a single factual question: a KB entry is a *complete set* of facts, so the model must decide not only what is true but how many objects, possibly none, a subject and relation admit.

The LM-KBC 2026 shared task (Alivanistos et al., 2022; Kalo et al., 2023, 2025) formalises this: given a subject s and relation r, predict the complete object set $\{o_1, \ldots, o_k\}$, where k may be zero, one, or many (Section 3). Predictions are scored by macro precision, recall, and F_1, with alias matching for entities and a 5% tolerance for numeric relations.

This is not single-answer probing. Unlike benchmarks such as LAMA (Petroni et al., 2019), which assume one gold object and reward ranking it highly, the task demands the *entire* set, and the empty set is a legitimate, frequent answer: a living person has no recorded city of death, and an island country has no land border. LM outputs are also stochastic, relations range from a single number to hundreds of recipients, and the model must recognise when it does not know an answer and *abstain* rather than fabricate one.

The 2026 edition is strictly *closed-book*: any open-weight model up to 32B parameters, but no web search, retrieval, external corpus, or fine-tuning. This forecloses the retrieval and tuning that strong entries in earlier editions relied on (Zhang et al., 2023; Asai et al., 2024), leaving parametric recall and prompt design as the only levers.

We start from the evaluation metric rather than the model. Because macro-F_1 scores an empty prediction against an empty gold set as perfect, predicting nothing everywhere already attains 0.169 relation-averaged macro-F_1 and outscores the official baseline on the most null-heavy relations (Section 3). Points therefore come not from answering everything, but from recalling the few numeric val-

[1] Repository link withheld for double-blind review; included in the camera-ready version.

ues and, above all, from knowing when *not* to answer. We make abstention a first-class decision: for each null-heavy relation we sample several times and keep an object only when the samples agree, collapsing to $\emptyset$ when the model is inconsistent, while scope-grounded prompts and median aggregation handle the rest (Figure 1).

This is a shared-task system paper making three claims of decreasing generality. First, a metric observation (Section 3): since a correct empty set scores a perfect F_1, the empty baseline already beats the official one on the null-heavy relations, so the task rewards *when to answer* as much as what to answer. This follows from the metric but is easy to miss, and it shapes our design. Second, a simple, reproducible pipeline (Section 4) with no trainable parameters, combining standard parts: scope-grounded prompting, self-consistency voting for abstention (a fixed a-priori θ=0.5), and median aggregation for numerics; on the null-heavy relations even predicting nothing beats the official baseline, and voting on top of the 14B adds a further, if modest, gain. Third, and most useful, an empirical finding: in our one controlled comparison at the $\leq$32B budget, a single-sample Gemma-4-31B (0.461/0.488, beating the official baseline on five of the six relations) outscores the smaller 4-bit Qwen2.5-14B even with self-consistency voting (0.416); whether voting also helps the 30B is untested, as API limits kept it off (Section 7).

2 Background and Related Work

Pretrained language models have long been probed as implicit knowledge bases, a line that Petroni et al. (2019) opened by casting factual recall as cloze probing. Subsequent work sharpened the prompts (Jiang et al., 2020; Shin et al., 2020; Zhong et al., 2021), quantified how much knowledge a model's parameters hold (Roberts et al., 2020), and questioned how reliably that knowledge can be read out (Cao et al., 2021; AlKhamissi et al., 2022). These probes share a common assumption: a single gold object per query, scored by how highly the model ranks it. We instead target the set-valued, alias-graded variant in which the empty set is a valid answer, a setting far closer to materialising an actual knowledge base than to ranking a single token (Pan et al., 2023).

A parallel thread turns language models toward building knowledge bases rather than merely probing them, and the LM-KBC shared task is its clearest instance. Across its editions (Alivanistos et al., 2022; Kalo et al., 2023, 2024, 2025) the task has moved from masked-LM probing and prompt ensembling toward instruction-tuned generative systems, with some earlier tracks permitting retrieval or fine-tuning (Zhang et al., 2023). The 2026 edition removes those options with its closed-book, no-tuning, $\leq$32B rules, and it is exactly this stripped-down regime that forces the parametric-recall-with-abstention design we pursue.

The usual route to knowledge-intensive accuracy under such pressure is retrieval augmentation (Lewis et al., 2020; Guu et al., 2020), which helps most precisely when parametric memory is weak (Mallen et al., 2023). Self-RAG (Asai et al., 2024) pushes this further by letting the model itself decide *when* to retrieve. We borrow the spirit of a model-driven, adaptive decision, but because retrieval is forbidden we redirect it to a different question: not what to fetch, but *when to answer at all*.

That decision draws together three further threads that converge in our method. Self-consistency (Wang et al., 2023; Chen et al., 2023) samples a model repeatedly and aggregates the generations to select a single answer; abstaining when uncertain is the classic reject option (Geifman and El-Yaniv, 2017), studied in NLP as calibration (Guo et al., 2017), unanswerable QA (Rajpurkar et al., 2018), and selective QA (Cole et al., 2023), and language models retain some awareness of what they know (Kadavath et al., 2022). We combine these: cross-sample *disagreement* (Kuhn et al., 2023; Manakul et al., 2023) serves as our zero-parameter confidence signal, and, unlike majority-vote self-consistency, which always emits an answer, we let that signal trigger *abstention*. Our setting differs from all of the above in one decisive way, and it is what makes abstention necessary rather than optional: the objects form a set of unbounded cardinality, the prediction is closed-book and untuned, and the empty set is itself a valid, high-value prediction rather than a failure to be avoided.

3 Task and Metric Analysis

The dataset comprises six Wikidata relations (Table 1) that fall into three groups with sharply different answer shapes. The *numeric* relations HASCAPACITY and HASAREA have a sin-

Relation	#val	empty%
countryLandBordersCountry	68	26
personHasCityOfDeath	100	39
companyTradesAtStockExchange	100	36
awardWonBy	10	0
hasCapacity	100	0
hasArea	100	0

Table 1: Validation statistics. awardWonBy averages ∼148 objects per subject (max 641); three string relations are often legitimately empty.

Algorithm 1 CalibratedAbstention

```
Require: samples S={s_1, ..., s_N}, threshold θ, multi
  c[o] ← #{s_i : o ∈ PARSE(s_i)} for each object o
  if multi then return {o : c[o]/N ≥ θ}
  else
      o* ← arg max_o c[o]; return c[o*]/N ≥ θ ? {o*} : ∅
  end if
```

gle scalar object; the *large-set* relation awardWonBy has up to hundreds of recipients per subject; and three *null-heavy* string relations, countryLandBordersCountry, personHasCityOfDeath, and companyTradesAtStockExchange, leave a large fraction of subjects with no valid object at all. Each relation carries a precise scope (land-only borders, city-granular death location, the highest published capacity), which we copy verbatim into the prompt so that the model targets the same notion the gold set was constructed under.

Predictions are scored against the gold object set by macro precision and recall. A predicted string counts as a true positive if its normalised form (lower-cased, with diacritics and punctuation removed) matches any alias of a gold object, where each gold object is given as a set of surface aliases; a numeric prediction counts if it lies within a 5% relative tolerance of the gold value. Two boundary conventions drive everything that follows: precision is defined as 1 when the prediction is empty, and recall as 1 when the gold set is empty. We report *relation-averaged* macro-F_1 (the mean of the six per-relation scores) and *micro-averaged* F_1 (over all instances).

These conventions follow directly from the metric definition, but their effect is easy to underestimate: because an empty prediction against an empty gold set is a perfect F_1, predicting the empty set for *every* validation instance already scores the "Empty" bars in Figure 3, averaging 0.169 relation-averaged macro-F_1 (0.195 micro). Although this trails the official baseline overall (0.300), on the two most null-heavy relations doing nothing already *beats* that baseline (personHasCityOfDeath 0.39 vs. 0.21; companyTradesAtStockExchange 0.36 vs. 0.35). The metric therefore rewards two narrow abilities rather than answering everything: recalling the numeric values, and abstaining precisely when no object exists. Our pipeline pulls exactly these two levers.

4 System

Overview. The pipeline (Figure 1) maps each subject–relation pair (s, r) to one chat prompt, decodes a cached set of samples (N for voting, n for the numeric median), and aggregates them per relation kind into an object set. It adds no trainable parameters of its own and targets the two analysis levers: abstention and numeric recall.

Scope-grounded prompts. Each relation's system message states the task, the *exact* ground-truth scope (copied verbatim from the official definitions), and the required answer form (Table 2); the model emits one JSON object `{"answers":[...]}` (Figure 2), which makes the empty case unambiguous. Few-shot exemplars come from the training split; for null-heavy relations we interleave empty and non-empty exemplars so abstention is demonstrated, not merely instructed. Even a 0.5B model emits valid JSON on 100% of instances, so format is never a bottleneck.

Calibrated abstention. Scope grounding tells the model *what* counts as an answer, but not *when* to stay silent. A bare instruction to "answer empty when unsure" is unreliable: left unconstrained, a small model names a death-city for living people and invents stock exchanges, dropping precision *below* the empty baseline (Section 7). Instead, for the three null-heavy relations we sample N=3 completions at temperature τ=0.7 and keep an object only if it appears in at least a fraction θ of samples; single-object relations abstain entirely if even the top answer is below θ (Algorithm 1). When the model guesses, samples disagree and the prediction collapses to $\emptyset$, exactly where the metric rewards silence. We disable voting for awardWonBy (recall-bound) and the numeric relations.

Numeric aggregation. The numeric relations need a different aggregator: hasCapacity and hasArea take a single number graded within 5%.

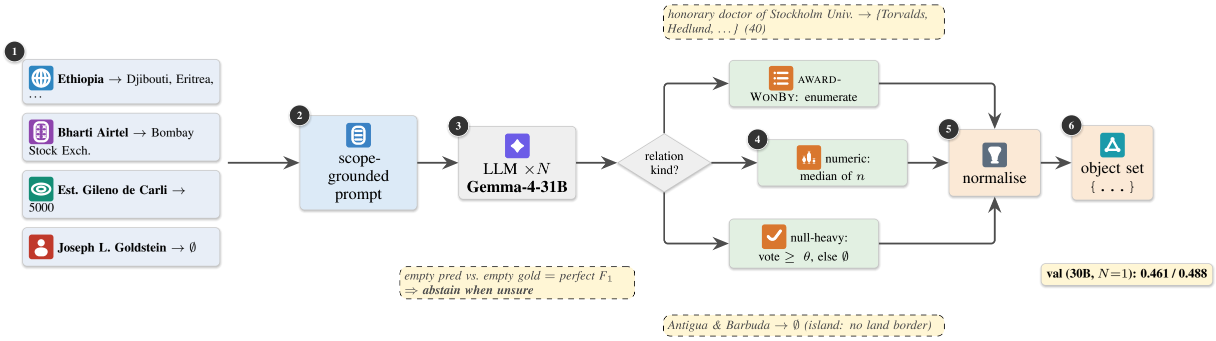

Figure 1: System pipeline (no trainable parameters). Real dataset instances (left, stage 1), shown with entity-type icons and their gold objects, are each rendered into a scope-grounded chat prompt (stage 2); the model is sampled N times (stage 3, cached), and the samples are aggregated by relation kind (stage 4): *median* for numeric relations, consistency *voting with abstention* (keep an object only if it appears in $\geq \theta$ of samples, else $\emptyset$) for the null-heavy string relations, and *enumeration* for AWARDWONBY; a final normalisation matches the evaluator exactly. Dashed callouts give worked examples, e.g. the island country ANTIGUA & BARBUDA correctly collapses to $\emptyset$; the input cards likewise include the still-living scientist JOSEPH L. GOLDSTEIN, mapped to $\emptyset$ (no city of death). The design follows from the metric insight (lower-left badge). The reported Gemma-4-31B score uses a single sample (N=1, voting off); the N-sample voting path is exercised by the 14B ablation (Section 7).

Relation	Scope stated in the prompt	Answer form
COUNTRYLANDBORDERSCOUNTRY	countries sharing a *land* border; maritime borders excluded; an island has none	countries, or []
PERSONHASCITYOFDEATH	the *city* where the subject died; empty if alive or the city is unknown	one city, or []
COMPANYTRADESATSTOCKEXCHANGE	stock exchange(s) on which the company is publicly listed; empty if unlisted	exchanges, or []
AWARDWONBY	every recipient of the award (often tens to hundreds)	names (long)
HASCAPACITY	maximum spectator capacity in people; highest published value	one integer
HASAREA	total surface area in km^2 (land + inland water)	one number

Table 2: Per-relation prompt specification. Each system message states the relation's exact scope (copied from the official definitions) and the required answer form; the model replies with a single JSON object `{"answers":[...]}`.

```
SYS: precise KB engine (closed book).
TASK: city where the subject died.
SCOPE: city granularity; [] if
alive/unknown.
Reply only {"answers":[...]}.
U: Egbert Mulder
A: {"answers":["Groningen"]}
U: Dave Keon
A: {"answers":[]}
U: <query>  A:
```

Figure 2: Abbreviated prompt for a null-heavy relation. Mixed exemplars teach abstention; output is forced JSON.

We sample n=5 completions and take the *median* of the parsed numbers (Wang et al., 2023), which resists outlier hallucinations.

Robust parsing. Whatever the relation kind, every sample passes through a common final stage: a relation-aware parser recovers the answer list from noisy output (code fences, prose), strips explicit empty markers, extracts one number for numeric relations (handling commas, units, scale words), and de-duplicates. Normalisation (lower-casing, diacritic/punctuation removal) is byte-identical to the official evaluator; pushing gold answers back through the parser scores 1.000, confirming it is lossless.

5 Experimental Setup

Models. Our primary system uses **Gemma-4-31B-it** (Gemma Team, 2026) (**Gemma-4-31B** for brevity), a 30.7B-parameter open-weight model within the 32B cap. It exceeds the memory of our single 16 GB GPU, so we serve it through a hosted OpenAI-compatible API. The weights are open and we apply no fine-tuning; the endpoint serves the unmodified weights as a plain text-completion service (no retrieval, tools, or augmentation), so it is compute-only and the closed-book rules hold. For the self-consistency ablations (Section 7) we additionally run **Qwen2.5-14B-Instruct** (Qwen Team, 2024) (14.7B) locally in 4-bit. Several nominally "32B" models in fact exceed the cap (Qwen2.5-32B at 32.5B, Qwen3-32B at 32.8B) and are ineligible.

Decoding. At decode time, the pipeline selects an aggregator per relation kind (Section 4): the median over n=5 samples for numeric relations, consistency voting over N=3 samples at threshold θ=0.5 for null-heavy relations (Algorithm 1), and a single greedy enumeration for AWARDWONBY. We evaluate these self-consistency configurations (temperature τ=0.7) on Qwen2.5-14B, which we can sample freely. API rate limits preclude multi-sampling Gemma-4-31B at scale, so we decode it with a *single* sample per query and confine the self-consistency ablations to the 14B (Section 6). Prompts are 8-shot for every relation except AWARDWONBY (2-shot), with exemplars from the training split (Brown et al., 2020).

Hardware. Qwen2.5-14B runs on a single NVIDIA T4 (16 GB) via HuggingFace Transformers in 4-bit (`bitsandbytes nf4`, `float16` compute) (Dettmers et al., 2023); quantisation only fits the model to the hardware and does not change the counted parameter budget. Gemma-4-31B is queried over its API from the same machine. Each model processes the full 478-instance validation set.

Baselines. We compare against two references on the same validation split: the *empty* baseline, which predicts $\emptyset$ for every instance (Section 3), and the *official* organiser baseline (a few-shot Qwen3.5-9B system) whose per-relation scores we take from the task repository.

Reproducibility. The pipeline caches every raw sample, so all post-processing (parsing, consistency voting, median aggregation, and the threshold sweep) is a deterministic, GPU-free function of that cache. Every ablation in Section 7 therefore comes from a single generation pass and reproduces without a GPU. We release all code and prompts.

6 Results

Our primary system, **Gemma-4-31B**, scores **0.461** relation-averaged macro-F_1 (**0.488** micro) on validation, versus 0.169/0.195 for the empty baseline, 0.300/0.313 for the official baseline, and 0.416/0.442 for the smaller Qwen2.5-14B. Figure 3 and Table 3 give the per-relation picture. Gemma-4-31B beats the official baseline on *five of the six* relations, the exception being HASCAPACITY (0.17 vs. 0.18, within decoding noise). Over the 14B, its gains concentrate where the model already has usable knowledge: the biggest 14B→30B jumps are on COMPANYTRADESATSTOCKEXCHANGE (0.61 → 0.74) and HASCAPACITY (0.09 → 0.17), while the near-saturated COUNTRYLANDBORDERSCOUNTRY reaches 0.971. It ties the 14B on HASAREA and trails it on PERSONHASCITYOFDEATH (0.42 vs. 0.45), the relation where the single-sample 30B forgoes the abstention voting the 14B uses. Strikingly, the 30B attains this overall lead *without* self-consistency or abstention, from a single sample. We could not run voting on the 30B, so we isolate the decode-time levers on the smaller models in Section 7.

Relation	Empty	Base	14B	**30B**
countryLandBorders	0.265	0.665	0.918	**0.971**
companyTrades	0.360	0.354	0.609	**0.737**
personCityOfDeath	0.390	0.210	**0.450**	0.420
hasArea	0.000	0.290	**0.330**	**0.330**
hasCapacity	0.000	**0.180**	0.090	0.170
awardWonBy	0.000	0.101	0.096	**0.140**
Avg. of relations	0.169	0.300	0.416	**0.461**
Micro (all inst.)	0.195	0.313	0.442	**0.488**

Table 3: Validation macro-F_1 by model. "Empty" is the measured empty baseline (predict nothing everywhere), "Base" the official baseline (Qwen3.5-9B); **30B** is our primary Gemma-4-31B system, 14B is Qwen2.5-14B. Best per row in bold. Gemma-4-31B beats the official baseline on five of the six relations (all but HASCAPACITY).

7 Analysis

Because the headline 30B result is single-sample, we isolate the decode-time levers on the 14B, which we can sample freely; all ablations use the same cached samples.

Abstention. Abstention matters most exactly where the metric rewards silence. On the three null-heavy relations, even predicting nothing beats the official 9B baseline (Section 3), so the question is whether a real model does better by abstaining *selectively* rather than never. It does: decoding the 14B with consistency voting lifts its null-heavy average from 0.642 (single sample) to 0.659

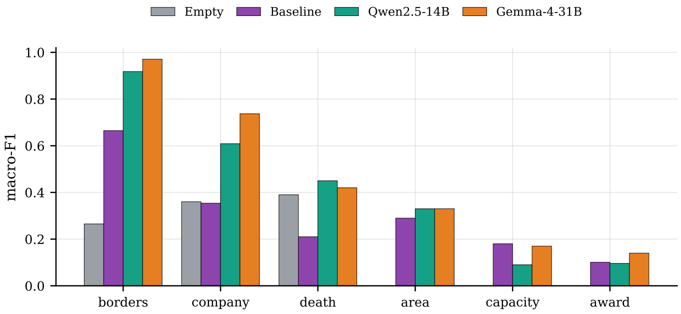

Figure 3: Per-relation validation macro-F$_1$: empty baseline, official baseline, Qwen2.5-14B, and Gemma-4-31B. Gemma-4-31B beats the official baseline on five of the six relations.

at θ=0.5. The gain is modest but consistent, and it comes from precision: raising θ collapses inconsistent (usually wrong) guesses to $\emptyset$, trading a little recall to suppress false positives.

Setting the threshold. How high should θ be? Figure 4 sweeps it for the 14B on the null-heavy relations. Our deployed systems fix θ=0.5; the sweep is diagnostic (run on the split we report on, so its optima are optimistic, not tuned back in). Even so, the pattern argues for per-relation thresholds: PersonHasCityOfDeath (39% empty) peaks at a low θ and falls off as voting grows stricter, while CountryLandBordersCountry and CompanyTradesAtStockExchange tolerate aggressive voting. Numeric relations are a different story: median aggregation is roughly neutral, so the gap there is knowledge-bound rather than variance-bound; the model's area and capacity guesses are often the right order of magnitude but fall outside the 5% band (the HasArea miss in Table 4).

The AwardWonBy recall cap. The relation AwardWonBy, which we beat the baseline on only narrowly, is recall-capped, and this limit constrains both models. Gold sets average ~146 recipients (max 610), but the 14B enumerates only ~11 names; its outputs are *not* truncated, it closes the JSON early, capping recall near $11/146 \approx 0.07$. The listed names are often plausible but wrong (P=0.39). Parametric recall and a reluctance to enumerate bound this relation; within a closed-book single model it is largely unsolved, and a larger in-budget model or multi-step decomposition would help most.

Qualitative behaviour. Table 4 shows representative Gemma-4-31B outputs: exact on well-known facts and near-complete on borders, with the two error modes above visible, a numeric near-miss on HasArea and plausible-but-wrong famous names on AwardWonBy.

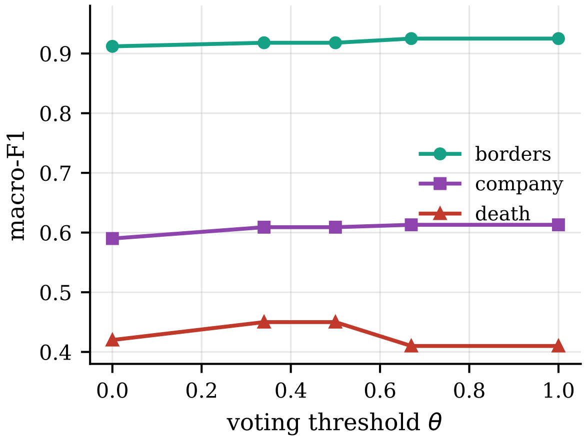

Figure 4: Per-relation validation F$_1$ vs. the voting threshold θ (Qwen2.5-14B) on the three null-heavy relations. The most null-heavy relation (Death, 39% empty) peaks at a low θ and declines as voting grows stricter, while borders and company tolerate aggressive voting, arguing for per-relation thresholds.

8 Conclusion

We treated *when to answer* as the central decision in closed-book, set-valued KB construction: because macro-F$_1$ rewards a correct empty set, abstention pays off, numeric recall is scarce, and scope grounding curbs over- and under-prediction. Instantiated with Gemma-4-31B, the system reaches 0.461/0.488 (relation-averaged/micro) and beats the official baseline on five of the six relations, reproducibly from a cached pass. The durable lesson is comparative: a single-sample 30B outscores a voted 14B, while voting mainly rescues weaker models, so pairing voting with the 30B is the clearest next step.

Relation	Subject	Gold (abbrev.)	Gemma-4-31B output	
borders	Ethiopia	Djibouti, Eritrea, Kenya, Somalia, S. Sudan, Sudan	Djibouti, Eritrea, Kenya, Somalia, S. Sudan	✓ 5/6
death	Vito Acconci	New York City	New York City	✓
company	Bharti Airtel	Bombay Stock Exch., NSE of India	Bombay Stock Exchange, NSE of India	✓
capacity	Gileno de Carli	5000	5000	✓
area	Lošinj	74.36	123.3	×
award	Stockholm hon. dr	Torvalds, Hedlund, Trebst, . . . (40)	Mandela, Tutu, Annan, Yunus, . . .	×

Table 4: Qualitative examples: Gemma-4-31B vs. gold. Exact hits on PERSONHASCITYOFDEATH/HASCAPACITY, near-complete borders; the numeric near-miss (HASAREA) and plausible-but-wrong enumeration (AWARDWONBY) are the knowledge-bound errors discussed above.

Limitations

Held-out evaluation. We report validation results only; no test-set numbers are available yet. Our reported systems use a fixed, a-priori θ=0.5, but the diagnostic threshold sweep (Figure 4) is run on the same split it is scored on, so its per-relation optima are optimistic rather than held-out and should be read as diagnostic only.

Decoding variance. We did not measure the run-to-run variance of the stochastic decoding (N=3, n=5), and AWARDWONBY has only 10 subjects. Several margins are within plausible decoding noise, so small deltas, such as the 0.03 gap between the 14B and 30B on PERSONHASCITYOFDEATH, are indicative only and should not be read as reliable per-relation orderings.

Model coverage. Being closed-book, the system inherits the base model's coverage and recency limits and cannot recover facts the model never stored. This caps AWARDWONBY recall and rare-venue numeric accuracy, and it means our gains reflect parametric recall rather than any external grounding.

Single-sample 30B. Because API rate limits precluded multi-sample decoding, our Gemma-4-31B scores are single-sample. We therefore cannot report how the 30B behaves under the self-consistency and abstention voting that help the smaller models; pairing them is left to future work and may further lift the null-heavy and numeric relations, so our 30B numbers are likely a floor rather than a ceiling.

Metric specificity. The abstention advantage stems from this task's macro metric and its empty-as-valid-answer setting. It may not transfer to single-answer or micro-only regimes, and practitioners porting the method to other benchmarks should re-verify that abstention is still a positive-value action before adopting the voting machinery.

Ethics and Generative-AI Declaration

Closed-book KB construction surfaces facts from model parameters that may be stale, biased, or hallucinated. Predictions about people (e.g. place of death) can be wrong, so they should not be treated as authoritative without verification. Our abstention mechanism partly mitigates this: when the model is uncertain it emits an empty answer rather than a confident fabrication. The only generative-AI component is the open-weight LM that produces the predictions, as the task requires; we used no proprietary assistant to generate results.

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